

Element Interactivity and Intrinsic, Extraneous, and Germane Cognitive Load

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Abstract In cognitive load theory, element interactivity has been used as the basic, defining mechanism of intrinsic cognitive load for many years. In this article, it is suggested that element interactivity underlies extraneous cognitive load as well. By defining extraneous cognitive load in terms of element interactivity, a distinct relation between intrinsic and extraneous cognitive load can be established based on whether element interactivity is essential to the task at hand or whether it is a function of instructional procedures. Furthermore, germane cognitive load can be defined in terms of intrinsic cognitive load, thus also associating germane cognitive load with element interactivity. An analysis of the consequences of explaining the various cognitive load effects in terms of element interactivity is carried out.

Keywords Cognitive load theory · Element interactivity · Extraneous cognitive load · Intrinsic cognitive load · Germane cognitive load

The allocation of working memory resources to deal with intrinsic cognitive load, concerned with the intrinsic complexity of information; extraneous cognitive load, concerned with the manner in which instruction is designed; and germane cognitive load, concerned with the acquisition of knowledge has been an important facet of cognitive load theory for some time. The mechanism underlying intrinsic cognitive load, element interactivity, has been specified for an even longer time. The mechanisms underlying extraneous and germane cognitive load have been specified in a differential manner for each task to which they are applied but tend not to be specified in unified theoretical terms (Beckmann 2010; Schnotz and Kurschner 2007) and that omission is beginning to result in some serious misunderstandings and contradictions concerning the relations between the categories of cognitive load. The following formulation is intended to address such issues by providing a uniform foundation for the division of cognitive load into categories. Intrinsic cognitive load and element interactivity will be discussed first.

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Intrinsic Cognitive Load and Element Interactivity

Intrinsic cognitive load (Sweller 1994; Sweller and Chandler 1994) is concerned with the natural complexity of information that must be understood and material that must be learned, unencumbered by instructional issues such as how the information should be presented or in what activities learners should engage to maximise learning. For a given task and given learner knowledge levels, it is fixed and cannot be altered other than by either changing the basic task or changing knowledge levels. Intrinsic cognitive load only can be altered by changing the nature of what is learned or by the act of learning itself.

The level of intrinsic cognitive load for a particular task and knowledge level is assumed to be determined by the level of element interactivity. An element is anything that needs to be or has been learned, such as a concept or a procedure. Low element interactivity materials allow individual elements to be learned with minimal reference to other elements and so impose a low working memory load. For example, learning chemical symbols or some of the nouns of a foreign language constitute low element interactivity tasks. Each element can be learned without reference to any other elements and so, although the task may be difficult because there may be many elements, it does not impose a heavy working memory load because individual elements can be learned independently of each other. For example, learning the chemical symbol for copper can be learned independently of learning the symbol for iron. Because they can be learned independently, working memory need only process the cognitive elements associated with the symbol for copper without the load associated with the symbol for iron.

High element interactivity material consists of elements that heavily interact and so cannot be learned in isolation. The more elements that interact, the heavier the working memory load. For example, when dealing with equations, all of the elements associated with an equation must be considered simultaneously because all of the elements interact. For a novice learning algebra and faced with the problem, $(a + b)/c = d$, solve for a , each of the symbols in the equation may act as an element, and all must be processed simultaneously in working memory for the equation to be understood. Learning how to solve this category of problems is a much higher element interactivity task than learning the chemical symbols of the periodic table. Although learning to solve simple algebra equations associated with the above problem is likely to be easier than learning the chemical symbols of the periodic table because there are far fewer elements that need to be dealt with, the equation is much higher in element interactivity and so will impose a much higher intrinsic cognitive load. The intrinsic cognitive load, in turn, has instructional implications that differ from the implications that flow from task difficulty associated with the total number of elements, rather than element interactivity. The instructional implications of element interactivity levels will be discussed in a subsequent section.

Element interactivity levels are determined by estimating the number of interacting elements (Sweller and Chandler 1994; Tindall-Ford *et al.* 1997). Such estimates must simultaneously take into account the nature of the information and the knowledge of learners. For example, for beginning readers, each letter read or even the constituent parts of letters may constitute an individual element. For readers of this article, combinations of words and phrases are more likely to constitute elements and when reading simple text, entire sentences may constitute elements.

Extraneous Cognitive Load and Element Interactivity

Working memory load is not only imposed by the intrinsic complexity of the material that needs to be learned. It also may be imposed by instructional procedures that are less than optimal. Nonoptimal instructional procedures are referred to as imposing an extraneous cognitive load. Cognitive load theory is primarily concerned with techniques designed to reduce extraneous cognitive load. Many such techniques have been devised and continue to be devised (Sweller 2003, 2004).

Although intrinsic cognitive load has used element interactivity as a common determiner of cognitive load irrespective of the nature of the information being dealt with, there has been little attempt to identify a similar underlying cause of extraneous cognitive load (Beckmann 2010). As a consequence, although there is a reasonably clear pattern to the generation of at least some of the identified cognitive load effects (for example, the expertise reversal effect followed from the redundancy effect that followed directly from the split-attention effect that, in turn, followed directly from the worked example effect), each effect has tended to be discussed in isolation from the other effects that have been identified. Each effect has been generated by and explained using cognitive load theory, but little attempt has been made to suggest that the various effects caused by an extraneous cognitive load might have a common underlying cause.

I suggest that element interactivity is the major source of working memory load underlying extraneous as well as intrinsic cognitive load. If element interactivity can be reduced without altering what is learned, the load is extraneous; if element interactivity only can be altered by altering what is learned, the load is intrinsic (Beckmann 2010). Of course, what constitutes intrinsic or extraneous cognitive load depends on what needs to be learned. For example, if the goal of learning is to comprehend concepts incorporated in some text, using jargon may constitute an extraneous cognitive load. Alternatively, if the goal is to learn the specialised language used in an area, the “jargon” is intrinsic to the task (Schnotz and Kurschner 2007). Consequently, the same information may impose an intrinsic or an extraneous cognitive load depending on what needs to be learned.

Thus, element interactivity not only underlies intrinsic but also extraneous and indirectly, as will be discussed below, germane cognitive load. If we consider the various sources of extraneous cognitive load such as those that lead to the goal-free, worked example, split-attention or redundancy effects, the instructional procedures that facilitate learning all seem to involve a reduction in elements that learners need to simultaneously process. Specifying which interacting elements are needed to process information presented in a particular manner appears straightforward for many cognitive load effects. If so, we have an underlying, explanatory mechanism for extraneous cognitive load that indicates why a particular instructional effect occurs. The element interactivity patterns associated with various cognitive load effects will be analysed in a subsequent section.

Germane Cognitive Load and Element Interactivity

Germane cognitive load also can be specified in terms of element interactivity, but the status of germane cognitive load differs from intrinsic and extraneous cognitive load. Intrinsic and extraneous cognitive load are determined by a combination of material and learner characteristics, but the emphasis is heavily on the characteristics of the material, in

that working memory load will be determined entirely by levels of element interactivity. The characteristics of the learner are only relevant in that what constitutes an element will depend on a learner's knowledge level. Multiple interacting elements for one learner with low knowledge levels may constitute a single element for a learner with a higher level of knowledge. For a given level of knowledge, element interactivity is solely determined by characteristics of the material.

In contrast to the emphasis by intrinsic and extraneous cognitive load on the characteristics of the material, germane cognitive load is concerned only with learner characteristics. It refers to the working memory resources that the learner devotes to dealing with the intrinsic cognitive load associated with the information. Except through its relation to the material via the interacting elements of intrinsic cognitive load, germane cognitive load is independent of the information presented. Importantly, it does not provide an independent source of working memory load. Assuming constant levels of motivation, the learner has no control over germane cognitive load. If intrinsic cognitive load is high and extraneous low, germane cognitive load will be high because the learner must devote a large proportion of working memory resources to dealing with the essential learning materials. If extraneous cognitive load is increased, germane cognitive load is reduced and learning is reduced because the learner is using working memory resources to deal with the extraneous elements imposed by the instructional procedure rather than the essential, intrinsic material. Thus, germane cognitive load is purely a function of the working memory resources devoted to the interacting elements that determine intrinsic cognitive load. The more working memory resources that must be devoted to extraneous cognitive load, the fewer will be available to deal with intrinsic cognitive load, reducing learning. This formulation assumes that motivation is high and all available working memory resources are being devoted to dealing with intrinsic and extraneous cognitive load.

Unlike intrinsic and extraneous cognitive load, germane cognitive load does not constitute an independent source of cognitive load. It merely refers to the working memory resources available to deal with the element interactivity associated with intrinsic cognitive load. If instruction is organised to allow working memory resources to deal primarily with the elements that impose an intrinsic cognitive load, germane cognitive load and so learning will be maximised. Instructional effectiveness will be compromised by the extent that instructional choices require learners to devote working memory resources to dealing with elements imposed by extraneous cognitive load. The more working memory resources are devoted to dealing with extraneous cognitive load, the less will be available to deal with intrinsic cognitive load and so the less will be devoted to germane cognitive load. On this formulation, as the element interactivity associated with extraneous cognitive load increases, subjective ratings of task difficulty (Paas and van Merriënboer 1993, 1994) or other measures of cognitive load such as secondary task measures (Chandler and Sweller 1996) should increase despite a decrease in germane cognitive load. A decrease in germane cognitive load means more working memory resources are being devoted to element activity associated with extraneous cognitive load, and fewer working memory resources are being devoted to element interactivity associated with intrinsic cognitive load. A decrease in extraneous cognitive load results in an increase in germane cognitive load as working memory resources are switched from elements associated with extraneous to elements associated with intrinsic cognitive load.

Advantages of This Formulation

In summary, a cognitive load can be imposed by instructional material and can be due to element interactivity associated with either intrinsic or extraneous cognitive load. Working memory resources must be allocated as far as possible to deal with those interacting elements. Working memory resources that deal with intrinsic cognitive load are germane to the task at hand and so are referred to as germane cognitive load. Working memory resources that deal with extraneous cognitive load are extraneous to the task at hand but must be dealt with if the instructional procedures demand those resources. Of course, there is no guarantee that sufficient resources will be available to deal with either intrinsic or extraneous cognitive load.

The advantage of this formulation is that it eliminates a potential contradiction. We know from countless experiments that we can detect changes in total cognitive load when intrinsic, extraneous, or germane cognitive load change. We also theorise that extraneous and germane cognitive load are complementary. When extraneous load goes down, we assume that germane load goes up. However, if germane load can compensate for extraneous load, why does total load change? It should remain constant but does not.

The contradiction is eliminated by the current formulation. Overall cognitive load is determined solely by an addition of intrinsic and extraneous cognitive load. If either changes but the other remains constant, total cognitive load changes. If extraneous cognitive load goes up, so will total cognitive load and that can be measured. Simultaneously, germane cognitive load goes down because more working memory resources must be devoted to handling extraneous cognitive load, and so, fewer are available to handle intrinsic cognitive load. The working memory resources devoted to handling intrinsic cognitive load are defined as germane cognitive load and decrease as extraneous cognitive load increases. Therefore, we can have complementarity between germane and extraneous cognitive load while total cognitive load varies.

Determining Cognitive Load

The three categories of cognitive load were introduced initially to explain otherwise inexplicable experimental results. For example, the extraneous cognitive load effects discussed below are obtainable using information that has a high intrinsic cognitive load but are reduced or eliminated when dealing with information that is low in intrinsic cognitive load (the element interactivity effect). This result is explained by assuming that if intrinsic cognitive load is low, sufficient working memory resources are available to deal with a higher extraneous cognitive load and so no extraneous cognitive load effects can be demonstrated. Without the concept of intrinsic cognitive load, such results would be difficult or impossible to explain. Nevertheless, although the distinction between intrinsic and extraneous cognitive load is required to explain experimental results, direct measures of individual categories of cognitive load can be problematic. For example, van Gog and Paas (2008) have discussed whether cognitive load should be measured during a learning or a test phase depending on the category of cognitive load.

The current formulation is designed to clarify some of the relevant issues. From an instructor's perspective, some interacting elements can be allocated to intrinsic cognitive load, whereas others can be allocated to extraneous cognitive load. In contrast, from a learner's perspective, all that is perceived is a number of interacting elements. Without

knowing the subject area or instructional design principles, the learner is not in a position to distinguish between those interacting elements due to extraneous or intrinsic cognitive load. Accordingly, attempts to independently distinguish between intrinsic and extraneous cognitive load using psychometric measures are likely to fail. The effects of these two categories of cognitive load can be independently predicted by analysing element interactivity for both categories prior to running an instructional experiment in which one of either intrinsic or extraneous cognitive load is kept constant while the other is varied. I doubt they can be psychometrically measured during an experiment if all that a learner experiences during learning is a given level of element interactivity. That level will reduce as learning occurs, but that reduction may not provide the learner with any indicators allowing a distinction between intrinsic and extraneous cognitive load. These two forms of cognitive load can readily be distinguished using a priori analyses of materials prepared for particular learners. Empirically, they can be distinguished experimentally by altering one form of cognitive load while keeping the other constant, but they probably cannot be distinguished psychometrically.

The current formulation, by closely relating both intrinsic and extraneous cognitive load to element interactivity, indicates why intrinsic and extraneous cognitive load are difficult or impossible to distinguish psychometrically. In addition, germane cognitive load also is likely to be difficult to distinguish psychometrically because it is defined in terms of the working memory resources required to deal with intrinsic cognitive load rather than as an independent entity exerting its own cognitive load. If germane cognitive load is defined in terms of working memory resources devoted to intrinsic cognitive load, the possibility of distinguishing between germane and intrinsic cognitive load by psychometric procedures is eliminated.

In contrast to the difficulty of determining the cognitive load that can be individually attributed to intrinsic, extraneous, and germane factors, overall cognitive load can be readily determined. Overall cognitive load, on the current formulation, consists of an addition of intrinsic and extraneous cognitive load. There is no reason why the currently commonly used subjective ratings of task difficulty or secondary task analysis (Brunken *et al.* 2003; 2004; Paas *et al.* 2003) cannot be used to determine changes in overall cognitive load. Whether those changes are due to alterations in intrinsic or extraneous cognitive load should be determined a priori by analysing element interactivity. A priori element interactivity analyses then can be used to provide hypotheses to be tested using conventional, randomised, controlled experiments. The effects discussed below were, without exception, determined using such experiments.

An analysis of element interactivity associated with various cognitive load effects

An analysis of element interactivity associated with various cognitive load effects dependent on extraneous cognitive load has not been carried out because previously, element interactivity only has been associated with intrinsic cognitive load. This section is concerned primarily with an analysis of element interactivity and extraneous cognitive load. In addition, extraneous cognitive load effects will be related to the much smaller number of cognitive load effects that in the past have been attributed to either intrinsic or germane cognitive load.

Goal-free effect

An early demonstration of this effect was provided by Sweller *et al.* (1983). The effect occurs when problem solvers learn less from solving conventional problems requiring them

to, for example, “find a value for x ” than goal-free problems requiring them to “find a value for as many variables as possible”. The effect occurs because when solving a conventional problem, problem solvers use a means-ends strategy (Newell and Simon 1972) that simultaneously requires them to consider the current problem state, the goal state, differences between the two states, problem-solving operators that can be used to reduce those differences, and any subgoals that have been established. Because each of these factors will themselves frequently involve several elements, the number of interacting elements can be overwhelming. In contrast, goal-free problems only require problem solvers to consider the current problem state and any operators that can be used to alter that state, resulting in a large decrease in interacting elements compared to conventional problems. Thus, the element interactivity associated with solving conventional problems by means-ends analysis should be much higher than solving the goal-free equivalents, allowing differential element interactivity to be used to explain the effect.

If we assume that learning to solve problems primarily consists of learning to recognise problem states and their associated moves, that is that working memory resources should be devoted to this activity, then learning to recognise problem states and their associated moves constitutes the intrinsic cognitive load associated with problem solving. Using a goal-free approach, activity is largely concerned with problem states and their associated moves. Working memory resources (germane load) should be devoted to this activity rather than to means-ends analysis (extraneous load) associated with conventional problems. A large number of experiments provide evidence for this hypothesis. (It should be noted that the effect only is likely to be obtainable in areas where finding the value of as many variables as possible only results in a limited number of variables being available. A large or infinite number of variables would render the technique impossible to use.)

Worked example effect

This effect occurs when students learn less from solving problems than from studying the equivalent worked examples (Cooper and Sweller 1987). When demonstrating the worked example effect, the control group is identical to the one used for the goal-free effect with learners required to solve a series of conventional problems. The worked example group is presented the same problems with their solutions provided.

As was the case for the goal-free effect, the problem-solving group must simultaneously consider a large number of interacting elements to use a means-ends strategy. The worked example group, when studying a worked example, merely has to look at each problem state and the move leading to the next state. The search through alternative problem states and alternative moves that is characteristic of problem solving and involves a large number of interacting elements is removed, reducing extraneous cognitive load by reducing element interactivity. The interacting elements associated with each problem state and its associated move remains when studying a worked example but that constitutes intrinsic cognitive load, providing no other extraneous elements have been added to the format of the worked examples leading to the split-attention or redundancy effects (see below). If working memory resources are devoted just to considering each problem state and its associated moves rather than a large range of possible moves, germane cognitive load is increased compared to problem solvers who must devote resources to the many interacting elements associated with possible but irrelevant moves. The worked example effect has probably been demonstrated more often than any other cognitive load effect.

Problem completion effect

This effect is closely related to the worked example effect (Paas and van Merriënboer 1994). Rather than providing a full worked example, a partially solved problem is presented with learners required to complete the solution. This technique is effective compared with full problems for the same reason that worked examples are effective. To the extent that a solution is provided, learners do not have to use a means-ends strategy and so the interacting elements associated with using the strategy to find those moves are eliminated from the elements that must be processed by working memory. Germane cognitive load and hence learning should be facilitated, a result that can be readily demonstrated empirically.

Split-attention effect

Instructional material is often presented in split-attention format, which increases extraneous cognitive load compared to physically integrated formats (Sweller *et al.* 1990). Assume a geometry worked example consisting of a diagram and a set of solution steps listed below the diagram. If one of the steps states, “Angle ABC = Angle XYZ”, learners need to engage in a search process to find the two angles. For Angle ABC, the interacting elements associated with that search consist of the statement “Angle ABC” and some, possibly most, of the angles of the figure. The learner must check each angle until the right one is found, and to prevent checking angles already checked, must attempt to remember previously checked angles. The same process must be engaged for Angle XYZ. Once both angles are found, the reason for their equality must be confirmed, but that confirmation involves intrinsic cognitive load, and so working memory resources devoted to those interacting elements constitute germane cognitive load.

The interacting elements associated with searching for the location of angles constitutes extraneous cognitive and should be eliminated. Eliminating the interacting elements involved in searching for angles can be accomplished by physically integrating the statement, “Angle ABC = Angle XYZ” with the diagram either by placing it in an appropriate location on the diagram rather than beneath the diagram or by using arrows to eliminate search. In that way, the learner only needs to consider the relevant angles rather than a range of angles, thus reducing element interactivity associated with extraneous cognitive load.

Redundancy effect

The redundancy effect occurs when unnecessary, additional information is presented to learners. If the same information is presented in diagrammatic and verbal form (Chandler and Sweller 1991) for example, learning may be facilitated by the elimination of the redundant verbal information assuming the content can be understood from the diagrammatic information alone. As an example, assume students are learning about the flow of blood in the heart lungs and body. They may be shown a diagram indicating that blood flows from the left ventricle to the body via the aorta. They also may be presented the statement, “blood flows from the left ventricle to the body via the aorta”. That statement is redundant.

In the above example, the diagrammatic and verbal elements interact. They do not need to interact, but the presentation mode ensures that they do interact. Learners are likely to search for correspondences between the diagram and the statements. Those correspondences require relating elements of information from the diagram and statements. The act of

coordinating these multiple, interacting elements is unrelated to learning the information being presented, imposing a heavy working memory load, because all the required information is presented in the diagram. The statements are redundant. Working memory resources devoted to dealing with the statements and their interaction with the diagram are unavailable for dealing with intrinsic cognitive load and so germane cognitive load is low. Eliminating the statements and so eliminating the extraneous, interacting elements increases germane cognitive load and facilitates learning.

Expertise reversal effect

This effect occurs when an instructional procedure “X” that is effective for novices in comparison to an alternative procedure “Y” becomes less effective as expertise increases (Kalyuga *et al.* 2003). With further increases in expertise, the alternative procedure “Y” may become more effective than “X”. There are many versions of the expertise reversal effect depending on which instructional effect is being investigated. All are ultimately dependent on the redundancy effect. For example, using novices, Kalyuga *et al.* (2001) obtained a conventional worked example effect with worked examples proving superior to solving the equivalent problems. With increasing expertise in the domain, that effect first disappeared and then reappeared as a reverse worked example effect. Solving problems was superior to studying worked examples for more expert learners. Studying worked examples is useful for novices in a domain. With increasing expertise, those worked examples merely demonstrate procedures that are already available to the learner from long-term memory, and so worked examples are redundant with further learning being facilitated by practicing solving problems.

Similarly, Kalyuga *et al.* (1998) obtained a conventional split-attention effect using novices with integrated text and diagrams proving superior to split-attention text and diagrams. With increasing expertise, textual information facilitated learning more by being eliminated rather than being integrated with diagrams. The textual information, essential for novices, was redundant for more knowledgeable learners.

The expertise reversal effect is dependent on a change in status of a set of interacting elements. Initially, for novices, a set of interacting elements reflect intrinsic cognitive load because they are essential to understanding. With increases in expertise, the same set of interacting elements reflect extraneous cognitive load because they are no longer needed. For a novice, studying problem solutions or reading text associated with diagrams may be important to understanding and learning in a particular domain. The interacting elements associated with that cognitive activity are intrinsic to learning and so constitute intrinsic cognitive load. Devoting working memory resources to those elements constitutes germane cognitive load. With increasing expertise, those same interacting elements are no longer needed. Rather than being intrinsic to learning, they are extraneous and should be eliminated. Thus, what constitutes intrinsic or extraneous cognitive load depends not just on the materials being used but also on the expertise of the learners. The same interacting elements may impose an intrinsic or extraneous load depending on levels of expertise.

Guidance fading effect

The worked example/problem-solving version of the expertise reversal effect leads directly to the guidance fading effect (Renkl and Atkinson 2003; Renkl *et al.* 2004). If novices need to be presented many worked examples whereas more expert learners should be presented problems, we might hypothesise that learners should first be presented worked examples,

followed by completion problems and then full problems. Work on the guidance fading effect has repeatedly demonstrated the advantages of this sequence.

The element interactivity profile associated with the guidance fading effect is identical to the profile associated with the expertise reversal effect. The interacting elements associated with worked examples are intrinsic to learning for novices. With increasing expertise, those intrinsic, interacting elements become extraneous to further learning because they are already part of long-term memory with learning being enhanced if the elements are eliminated.

Isolated-interacting elements effect

This effect relies on changes to intrinsic rather than extraneous cognitive load. Under some conditions, element interactivity that contributes to intrinsic cognitive load is so high that the interacting elements cannot possibly be processed simultaneously by working memory because they exceed working memory capacity. Such material only can be learned by treating the interacting elements as though they are isolated, that is by learning each element as though it is unrelated to the other elements with which it interacts. Subsequently, once learned as individual elements, the interactions between elements also can be learned without overloading working memory. It follows, that if interacting elements are presented initially as isolated elements to be learned and subsequently presented as fully interacting elements, learning may be facilitated compared to conditions in which learners only are exposed to fully interacting elements for the same amount of time. This result was obtained by Pollock *et al.* (2002).

The isolated-interacting elements effect depends on artificially altering the element interactivity associated with intrinsic cognitive load. The intrinsic cognitive load of a task cannot be changed but the task itself can be changed to a different task. The isolated-interacting elements effect relies on initially changing the task required of learners. Instead of attempting to learn how a set of elements interact, normally the main goal of a task that includes interacting elements, students initially learn the individual elements without learning how they interact. Unless learners subsequently also learn how elements interact, they are likely to be severely limited in what problems they can solve because the more important act of learning how the elements interact has been left until after the individual elements have been learned, two quite different tasks. Nevertheless, by delaying learning how elements interact, working memory is freed, facilitating learning compared to students required to both learn individual elements and how they interact, simultaneously.

Molar-modular effect

An example of this effect may be found in Gerjets *et al.* (2006). The basic rationale is similar to the isolated-interacting elements effect with the molar-modular effect depending on changing the intrinsic nature of what is to be learned to reduce intrinsic cognitive load. In the case of molar worked examples, learners are presented a fully integrated technique for solving a particular category of problems. This approach results in many interacting elements that need to be processed simultaneously. The modular technique presents a different approach that divides the solution into separate modules with the element interactivity and hence intrinsic cognitive load substantially reduced. Experimental results indicated that the modular procedure resulted in more learning and superior problem solving on subsequent tests than the molar approach.

The technique differs from the isolated-interacting elements effect in two respects. First, in the examples demonstrating the effect, the material taught to the “molar” group is quite different from the material taught to the “modular” group. For example, Gerjets *et al.* taught learners how to solve probability problems either by using an equation (molar condition) or by using a logical series of modular steps (modular condition). These two conditions differ in many ways, whereas the isolated-interacting elements condition varies only in that the interacting elements condition, while including all of the information of the isolated condition, also has additional information concerning the relations between elements.

The second difference between the isolated-interacting elements effect and the molar-modular effect is that the modular condition, when demonstrating the molar-modular effect, is not exposed to the molar condition as well. In contrast, when demonstrating the isolated-interacting elements effect, the isolated condition is exposed to isolated elements first followed by interacting elements while the interacting elements condition is given instruction that equals the isolated elements condition in length but consists solely of interacting elements. A molar followed by modular condition is not used to demonstrate the molar-modular effect. Despite these methodological differences, the rationale for both effects is identical with both effects assuming that reducing element interactivity due to intrinsic cognitive load can facilitate learning of very high element interactivity material that otherwise can be difficult to process.

Variability effect

Under the current formulation, this effect also is due to alterations in element interactivity due to intrinsic cognitive load although the effect has previously been explained solely in terms of variations in germane cognitive load. Paas and Van Merriënboer (1994) tested the effects of teaching a procedure using worked examples that were similar in surface structure as opposed to worked examples with a greater variation in structural features. The results indicated that transfer performance was improved by a greater variation in structural features. Increased variability also was associated with increased cognitive load.

These results can be explained by changes in intrinsic cognitive load. Learners not only must learn how to solve particular categories of problems, but they also must learn which categories of problems require similar solutions. In low variability conditions, learners are simply taught how to solve a class of problems. In high variability conditions, learners are not only taught how to solve the class of problems, they are also taught some of the various categories to which the solutions apply. The elements that generate the differences between categories must be compared and learned. Element interactivity under high variability conditions is likely to be substantially higher than under low variability conditions because under low variability conditions, a similar set of surface elements is associated with each problem reducing the number of elements that must be dealt with.

The increase in element interactivity associated with high variability problems will be associated with an increase in germane cognitive load under the current formulation. It is important for students to learn to recognise which problems belong to a particular category, high variability worked examples will demonstrate the relevant problems and more working memory resources will be required to deal with the increased number of interacting elements compared to low variability problems.

The last three effects discussed alterations in intrinsic cognitive load rather than the extraneous cognitive load of the other effects. Cognitive load theory was primarily designed to account for changes in extraneous cognitive load. When dealing with extraneous cognitive load, recommendations tend to be simple and straightforward—extraneous

cognitive load should always be reduced because a reduction in extraneous cognitive load will have either a beneficial or at worst, neutral effect. When dealing with intrinsic cognitive load, whether the load should be increased or decreased is more complicated. If a subject area is to be learned, intrinsic cognitive load is inevitable. If it is too high to be handled by working memory, it must be reduced even if a temporary failure to fully understand essential concepts occurs. The instructional recommendations associated with the isolated-interacting elements and molar-modular effect result. Element interactivity must be reduced if working memory cannot handle all of the elements.

In contrast, if an instructional procedure has a low intrinsic cognitive load and learners fail to learn to process essential interacting elements, intrinsic cognitive load must be increased, providing the increase does not exceed working memory capacity. The variability effect is an example of the need to increase intrinsic cognitive load to include interacting elements important to a task. Of course, we can predict that if the intrinsic cognitive load is increased too much and too rapidly by increased variability, resulting in too many interacting elements for working memory to handle, then element interactivity due to variability would need to be reduced. Determining the appropriate level of intrinsic cognitive load is likely to be much more difficult than determining techniques for altering extraneous cognitive load because extraneous cognitive load always should be reduced. In contrast, intrinsic cognitive load has an optimal level with increases or decreases about that level both having negative consequences. Indeed, in an informal sense, determining the appropriate level of intrinsic cognitive load is a problem that all instructors routinely face when determining how much information to provide students.

Element interactivity effect

This effect refers to the fact that if intrinsic cognitive load is low, other cognitive load effects cannot be obtained. Cognitive load effects only can be obtained if intrinsic cognitive load is high (Sweller and Chandler 1994). Under the current formulation, the issue resolves down to the total level of element interactivity. If the element interactivity due to intrinsic cognitive load is low, when, for example students are learning a new scientific vocabulary in which each new item can be learned independently from all other items, the manner in which the material is presented may not matter a great deal. If the manner of presentation results in relatively high levels of element interactivity due to extraneous cognitive load, the working memory load may still not be sufficiently high to exceed working memory capacity if intrinsic element interactivity is very low. Altering the instructional procedures to reduce extraneous cognitive load may have minimal effects on learning effectiveness because the total cognitive load may be far below working memory capacity.

Alternatively, if element interactivity due to a combination of intrinsic and extraneous factors is high, reducing the element interactivity due to extraneous factors may be important. If an instructional format involves many interacting elements as occurs under split-attention or redundancy conditions and if those interacting elements are added to the intrinsic interacting elements associated with learning, for example to manipulate equations, the total number of interacting elements and so the total cognitive load may vastly exceed working memory capacity. Reducing the number of interacting elements associated with extraneous cognitive load can reduce working memory load to manageable proportions. Thus, if element interactivity due to intrinsic cognitive load is high, reducing the element interactivity due to extraneous cognitive load may be critical. If intrinsic cognitive load is low, reducing extraneous cognitive load may have little effect because the total cognitive load due to element interactivity may be less than working memory capacity.

On this formulation, the element interactivity effect is based on the combined element interactivity associated with both intrinsic and extraneous cognitive load. As such, it is central to cognitive load theory and all cognitive load effects. It needs to be treated as a separate effect nevertheless because there is considerable empirical evidence that variations in element interactivity due to intrinsic cognitive load can have profound effects on other cognitive load effects dependent on extraneous cognitive load (Sweller and Chandler 1994; Tindall-Ford *et al.* 1997). Adding the interacting elements associated with a high extraneous cognitive load to the interacting elements associated with a high intrinsic cognitive load can exceed working memory capacity. Adding the same interacting elements associated with a high extraneous cognitive load to a low intrinsic cognitive load may not matter a great deal if it does not exceed working memory capacity.

Modality effect

This effect occurs when multiple sources of information that cannot be understood in isolation and so must be processed simultaneously are presented using spoken rather than written text (Tindall-Ford *et al.* 1997). A control group consists of conventionally presented, split-attention information. A diagram and written or spoken text provide typical examples, but the effect can be obtained using any two or more sources of information with the above characteristics.

The modality effect differs from all of the other effects discussed in that it is not due to alterations in element interactivity. Although, like all cognitive load theory effects, it only can be obtained under conditions of high intrinsic cognitive load and thus high element interactivity, the effect is not caused by differences in element interactivity between conditions. Rather it is caused by a basic characteristic of the human cognitive system: Working memory can be expanded by using both auditory and visual processors. It is this expansion of working memory capacity that generates the benefits of using dual-modality presentations rather than a reduction of element interactivity.

Imagination effect

If learners are asked to imagine a procedure or concept rather than study information describing the procedure or concept, under some circumstances, learning can be facilitated (Leahy and Sweller 2005). The effect is explained under the current formulation by assuming that by imagining a procedure or concept, working memory resources are directed towards processing the interacting elements associated with intrinsic cognitive load rather than processing other elements that are extraneous to learning. In other words, germane cognitive load is increased because working memory resources devoted to intrinsic cognitive load are maximised. Simultaneously, and as a consequence, working memory resources devoted to other, extraneous aspects of the task are reduced.

The imagination effect has many of the characteristics of other cognitive load effects discussed above. Apart from only occurring under conditions of high element interactivity, the expertise reversal effect can be demonstrated using imagination instructions. The effect only occurs when learners are able to process interacting elements in working memory and that ability requires sufficient levels of expertise to enable the requisite processing. Until those levels are attained, studying is superior to imagining. Thus, for novices, imagining acts as an extraneous cognitive load while for more expert learners, studying material that no longer requires study acts as an extraneous cognitive load.

Self-explanation effect

There are other effects that have been analysed using cognitive load theory, such as the self-explanation effect (Chi *et al.* 1989; Renkl 1997). Under the current formulation, the self-explanation effect can be treated in the same way as the imagination effect, to which it may be related. Self-explanation instructions require learners to self-explain new procedures or concepts. The act of self-explanation directs cognitive resources (germane cognitive load) to deal with relevant interacting elements (intrinsic cognitive load). Engaging in other activities unrelated to the relevant interacting elements constitutes an extraneous cognitive load that interferes with learning. This explanation is identical to the one used for the imagination effect suggesting that instructions to imagine and instructions to self-explain may be similar.

It needs to be noted that the rationale for the imagination and self-explanation effects differ somewhat from the rationale for other effects associated with extraneous activities. Extraneous cognitive load usually is reduced by altering the instructional materials to reduce the working memory resources devoted to activities irrelevant to learning. In the case of the imagination and self-explanation effects, the materials are not altered. Both effects rely on a reduction in extraneous cognitive load, not by altering the materials but by encouraging learners to engage in activities different to their usual learning activities. Those usual activities increase extraneous cognitive load by misdirecting working memory resources away from the interacting elements associated with the information. Imagination or self-explanation instructions redirect working memory resources to the interacting elements that constitute the core of the information and in this manner reduce extraneous cognitive load.

Thus, there are two general techniques for reducing extraneous cognitive load. We can alter the instructional materials so that learners no longer engage in activities extraneous to learning or we can directly instruct learners to use cognitive processes that encourage them, rather than an instructor, to eliminate activities extraneous to learning by engaging in activities conducive to learning.

Conclusions

Element interactivity has been an important facet of cognitive load theory since Sweller (1994) and Sweller and Chandler (1994). The current formulation suggests it should be central to the theory. Whereas previously, element interactivity was restricted to providing a common explanatory mechanism for intrinsic cognitive load irrespective of the material being learned, the current work suggests it also can provide an explanatory mechanism for extraneous cognitive load irrespective of the instructional cause of that load. Accordingly, most cognitive load effects, whether based on variations of intrinsic or extraneous cognitive load, may be explainable using the common concept of element interactivity.

There may be considerable theoretical and practical advantages to the current formulation. From a theoretical perspective, relations between intrinsic, extraneous, and germane cognitive load are provided with a more rational foundation. Intrinsic cognitive load is caused by element interactivity that cannot be eliminated without altering the nature of the material that is learned. Extraneous cognitive load is element interactivity that is caused by instructional factors and so can be eliminated by altering instructional procedures. Germane cognitive load belongs to a quite different category to intrinsic and extraneous cognitive load. It simply consists of working memory resources used to handle

element interactivity associated with intrinsic cognitive load. As such, it is not an independent source of cognitive load like intrinsic and extraneous cognitive load. Total cognitive load is determined by total element interactivity generated by intrinsic and extraneous cognitive load.

Subjective rating scales, introduced to cognitive load theory by Paas (1992), can be used to determine overall cognitive load. If, for example, intrinsic cognitive load remains constant, and extraneous cognitive load is varied by varying instructional techniques, then any change in cognitive load obtained by using a subjective rating scale can be attributed to changes in extraneous cognitive load. Considerable evidence for changes in extraneous cognitive load has been accumulated using this technique. In contrast, evidence for any differences in intrinsic and extraneous cognitive load using psychometric techniques not associated with randomised experimental designs is largely absent. Because such evidence would require novices to determine whether element interactivity was associated with intrinsic or extraneous cognitive load, a relatively simple task for instructors but a difficult task for novice learners, psychometric evidence for a distinction between intrinsic and extraneous cognitive load may not become available.

The major practical advantage of the current formulation is that it could lead to the generation of further instructional effects. By defining both intrinsic and extraneous cognitive load in terms of element interactivity, it may be possible to analyse element interactivity prior to an experiment and so more easily predict experimental outcomes. Since its inception, the primary goal of cognitive load theory has been the generation of instructional procedures. Hopefully, the current formulation will contribute to this goal.

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